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Graph Theory & Combinatorics

1. Permutations and Combinations

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1.1 Basic Counting Principle

- Set S, the number of objects in it, denoted by $|S|$, sometimes called size

Addition Principle.
Suppose that set S is partitioned into pairwise disjoint parts $S_1, S_2, \ldots, S_n$. Then the size of S can be determined by finding the number of objects in each of the parts, and adding the numbers so obtained: $|S| = |S_1| + |S_2| + \ldots + |S_n|$.

Example. Let S be the set of students attending the combinatorics lecture. It can be partitioned into parts S_1 and S_2 where

$$S_1 = \text{set of students that } \textit{like} \text{ easy examples.}$$
$$S_2 = \text{set of students that } \textit{don't like} \text{ easy examples.}$$

If $|S_1| = 22$ and $|S_2| = 8$ then we can conclude $|S| = 30$.

Multiply Principle.
Let S be the set of ordered pairs (a, b), where the first object a comes from a set of size p, and for each choice of object a there are q choices for object b. Then the size of S is $p \times q$, i.e., $|S| = p \times q$.

Example. Consider the license plates in Karlsruhe. They have the form

$$\texttt{KA} - l_1 l_2 \; n_1 n_2 n_3$$

where l_1, l_2 are letters from $\{A, \ldots, Z\}$ and n_1, n_2, n_3 are digits from $\{0, \ldots, 9\}$. However, l_2 and n_3 may be omitted.

We count the number of possibilities for each position:

K	A	-	l_1	l_2	n_1	n_2	n_3
↑	↑	↑	↑	↑	↑	↑	↑
1	1	1	26	27	10	10	11

Here we modeled the case that l_2 is omitted by allowing an additional value $\perp$ for l_2 that stands for "omitted". The same holds for n_3.

By the Multiplication Principle the size of this set is given as $1 \cdot 1 \cdot 1 \cdot 26 \cdot 27 \cdot 10 \cdot 10 \cdot 11 = 772200$.

Subtraction Principle.

Let A be a set and let U be a larger set containing A. Let

$$\bar{A} = U \backslash A = \{x \in U : x \notin A\}$$

be the complement of A in U. Then the number of $|A|$ of objects in A is given by the rule

$$|A| = |U| - \bar{A}.$$

Example. If T is the set of students studying at KIT and S the set of students studying neither math nor computer science. If we know $|T| = 23905$ and $|S| = 20178$, then we can compute the number $|\overline{S}|$ of students studying either math or computer science:

$$|\overline{S}| = |T| - |S| = 23905 - 20178 = 3727.$$

Division Principle.

Let S be a finite set that is partitioned into k parts in such a way that each part contains the same number of objects. Then the number of parts in the partition is given by the rule

$$k = \frac{|S|}{number\ of\ objects\ in\ a\ part}.$$

Bijection Principle.
If S and T are finite sets, then $|S| = |T|$ if and only if there exists a bijection between S and T.

Example. Let S be the set of students attending the lecture and T the set of homework submissions for the first problem sheet.

If the number of students and the number of submissions coincide, then there is a a bijection between students and submissions.[1] and vice versa.

Note that in this and the above principles, we do not allow infinite sets. The reason for this is that infinite sets have some counter-intuitive properties regarding their size.

There is for example a bijection between the set $\mathbb{N}$ of all natural numbers and the set $2 \cdot \mathbb{N}$ of all even natural numbers. It is given by mapping $x \mapsto 2x$. In that sense, there are just as many natural numbers as there are even natural numbers, despite the fact that an infinite set of numbers is natural but not even.

[1] Whether a "natural" correspondence can easily be determined is an entirely different matter: That would require that you *cleanly* write *your name* onto your submission.

1.1 Basic Counting Principle

Pigeonhole Principle.

Let $S_1, \ldots, S_m$ be finite sets that are pairwise disjoint and $\sum_{i=1}^{m} |S_i| = n$. Then

$$\exists i \in \{1,\ldots,m\} : |S_i| \geq \lceil \frac{n}{m} \rceil \text{ and } \exists j \in \{1,\ldots,m\} : |S_j| \leq \lfloor \frac{n}{m} \rfloor$$

Example. Assume there are 5 holes in the wall where pigeons nest. Say there is a set S_i of pigeons nesting in hole i. Assume there are $n = 17$ pigeons in total. Then we know:

- There is some hole with at least $\lceil 17/5 \rceil = 4$ pigeons.
- There is some hole with at most $\lfloor 17/5 \rfloor = 3$ pigeons.

Double counting

If we count the same quantity in two different ways, then this gives us a (perhaps non-trivial) identity.

Example (Handshaking Lemma)**.** Assume there are n people at a party and everybody will shake hands with everybody else. How many handshakes will occur? We count this number in two ways:

First way: Every person shakes $n-1$ hands and there are n people. However, two people are involved in a handshake so if we just multiply $n \cdot (n-1)$, then every handshake is counted twice. The total number of handshakes is therefore $\frac{n \cdot (n-1)}{2}$.

Second way: We number the people from 1 to n. To avoid counting a handshake twice, we count for person i only the handshakes with persons of lower numbers. Then the total number of handshakes is:

$$\sum_{i=1}^{n}(i-1) = \sum_{i=0}^{n-1} i = \sum_{i=1}^{n-1} i.$$

The identity we obtain is therefore $\sum_{i=1}^{n-1} i = \frac{n \cdot (n-1)}{2}$.

- Define $[n] := \{1, \ldots, n\}$ to be the set of the first n natural number and let X be a finite set.
- An ordered arrangement of n elements of X is a map $s : [n] \to X$.
- Ordered arrangements are so common that they occur in many situations and are known under different names. We might be inclined to say that "Banana" is a string (or word) and that "0815422372" is a sequence even though they are both ordered arrangements of characters. Depending on the circumstances different notation may be in order as well.

We will introduce the most common perspectives on what an ordered arrangement is and the accompanying names and notation. Throughout, we use the example $X = \{\square, \bigcirc, \mid, \triangle, \nabla\}$ and the ordered arrangement s of $n = 7$ elements given as the table:

i	1	2	3	4	5	6	7
s(i)	$\square$	$\bigcirc$	$\bigcirc$	$\triangle$	$\bigcirc$	∇	∇

function: We call $[n]$ the *domain* of s and $s(i)$ the *image* of i under s ($i \in [n]$). The set $\{x \in X \mid s(i) = x \text{ for some } i \in [n]\}$ is the *range* of s.

In the example, the domain of s is $\{1, 2, 3, 4, 5, 6, 7\}$, the image of 3 is $\bigcirc$ and the range of s is $\{\square, \bigcirc, \triangle, \triangledown\}$. Note that $\shortmid$ is not in the range of s.

string: We call s an *X-string* of *length* n and write $s = s(1)s(2)s(3)\cdots s(n)$. The i-th position (or *character*) in s is denoted by $s_i = s(i)$. The set X is an *alphabet* and its elements are *letters*. Often s is called a *word*.

In the example, we would say that $s = \square\bigcirc\bigcirc\triangle\bigcirc\triangledown\triangledown$ is a string (or word) of length n over the five-letter alphabet X. The fourth character of s is $s_4 = \triangle$.

tuple: We can view s as an element of the n-fold *Cartesian product* $X_1 \times \ldots \times X_n$, where $X_i = X$ for $i \in [n]$. We call s a *tuple* and write it as $(s_1, s_2, \ldots, s_n)$. The element s_i is called the i-th *coordinate* ($i \in [n]$). Viewing arrangements as elements of products makes it easy to restrict the number of allowed values for a particular coordinate (just choose $X_i \subsetneq X$).

In the example we would write $s = (\square, \bigcirc, \bigcirc, \triangle, \bigcirc, \triangledown, \triangledown)$. Its first coordinate is $\square$.

sequence: If X is a set of numbers, then s is often called a *sequence* and s_i is the i-th *position*, *term* or *element* in the sequence. We would not use this terminology in the example above but could have, for instance, $s(i) := 3i + 2/5$. In that case the fourth term is equal to 12.4.

In the following we will mostly view arrangements as functions but will freely switch perspective when appropriate.

The most important ordered arrangements are those in which the mapping is injective, i.e., $s(i) \neq s(j)$ for $i \neq j$.

Definition 1.1 (Permutation)**.** Let X be a finite set. We define

permutation: A *permutation* of X is a bijective map $\pi : [n] \to X$. Usually we choose $X = [n]$ and denote the set of all permutations of $[n]$ by S_n.

We tend to write permutations as strings if $n < 10$, take for example $\pi = 2713546$ by which we mean the function:

i	1	2	3	4	5	6	7
$\pi(i)$	2	7	1	3	5	4	6

1.2.1 Permutations

Let k be a positive integer.

Permutation.

An k-permutation of a set S of n elements, is an ordered arrangement of k of the n elements, i.e., an injective map $\pi : [k] \to S$. The set of all k-permutations is denoted by $P(n, k)$, ($P(n, k) = n^{\underline{k}}$, pronounced "$n$ to the k falling").

Theorem

For n and k positive integers with $k \leq n$,

$$P(n, k) = n \times (n-1) \times \ldots \times (n-k+1) = \prod_{i=0}^{k-1} (n-i).$$

- $n^{\overline{k}} = n \times (n+1) \times \ldots \times (n+k-1) = \prod_{i=0}^{k-1} (n+i)$, pronounced "$n$ to the k raising"
- If $k > n$ then $P(n, k) = 0$
- Clearly, $P(n,1) = n$ for each positive integer n
- $P(n,0) = 1$

1.2.1 Permutations

- All permutations we have just considered are called linear permutations: the objects are arranged in a line.
- If instead of arranging objects in a line, arrange them in a circle.
- Suppose six children are marching a a circle
  - How many different ways can they form their circle?
  - Their positions relative to each other.
- Two circular permutations are the same if one can be brought to the other by a rotation, i.e., by a circular shift.
- There are six linear permutations for each circular permutation

Circular k-Permutation: We say that two k-permutations $\pi_1, \pi_2 \in P(n,k)$ are *circular equivalents* if there exists a shift $s \in [k]$ such that the following implication holds:

$$i + s \equiv j \pmod{k} \quad \Rightarrow \quad \pi_1(i) = \pi_2(j)$$

This equivalence relation partitions $P(n,k)$ into equivalence classes. A class is called a *circular k-permutation*. The set of all circular k-permutations is denoted by $P_c(n,k)$.

Take for example $\pi_1 = 76123$, $\pi_2 = 12376$ and $\pi_3 = 32167$. Then π_1 and π_2 are circular equivalents as witnessed by the shift $s = 3$. They are therefore representatives of the same circular 5-permutation (with elements from [7]). π_3 belongs to a different circular 5-permutation. Consider Figure 8 for a visualization.

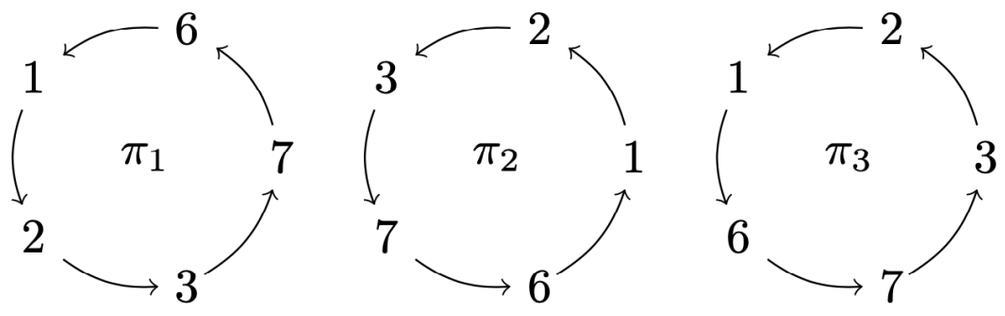


Figure 8: Visualization of the 5-permutations π_1, π_2 and π_3 where values are arranged in a circular fashion (the first value being on the right of the cycle). Note that π_1 and π_2 can be transformed into one another by "turning". They are therefore circular equivalents. Flipping is not allowed however, so π_3 is not equivalent to π_1 and π_2.

1.2.1 Permutations

We now count the number of (circular) k-permutations. For this we need the notation of factorials: $n! := 1 \cdot 2 \cdot \dots \cdot n$. Note that 0! is defined to be 1.

Theorem 1.2. *For any natural numbers $0 \leq k \leq n$ we have*

(i) $|P(n,k)| = n \cdot (n-1) \cdot \ldots \cdot (n-k+1) = \frac{n!}{(n-k)!}$.

(ii) $|P_c(n,k)| = \frac{n!}{k \cdot (n-k)!}$.

Proof. (i) A permutation is an injective map $\pi : [k] \to [n]$. We count the number of ways to pick such a map, picking the images one after the other. There are n ways to choose $\pi(1)$. Given a value for $\pi(1)$, there are $n-1$ ways to choose $\pi(2)$ (we may not choose $\pi(1)$ again). Continuing like this, there are $n-i+1$ ways to pick $\pi(i)$ and the last value we pick is $\pi(k)$ with $n-k+1$ possibilities. Every k-permutation can be constructed like this in exactly one way.

The total number of k-permutations is therefore given as the product:

$$|P(n,k)| = n \cdot (n-1) \cdot \ldots \cdot (n-k+1) = \frac{n!}{(n-k)!}.$$

(ii) We doubly count $P(n,k)$:

First way: $|P(n,k)| = \frac{n!}{(n-k)!}$ which we proved in (i).

Second way: $|P(n,k)| = |P_c(n,k)| \cdot k$ because every equivalence class in $P_c(n,k)$ contains k permutations from $P(n,k)$ (since there are k ways to rotate a k-permutation).

From this we get $\frac{n!}{(n-k)!} = |P_c(n,k)| \cdot k$ which implies the claim. □

- Let X be a finite set and $k \in N$.
- An unordered arrangement of k elements of X is a multiset S of size k with elements from X.

Take for example $X = \{\square, \bigcirc, \mid, \triangle, \nabla\}$, then an unordered arrangement of 7 elements could be $S = \{\square, \square, \bigcirc, \triangle, \triangle, \triangle, \nabla\}$. Order in sets and multisets does not matter, so we could write the same multiset as $S = \{\triangle, \square, \triangle, \nabla, \square, \triangle, \bigcirc\}$.

However, we prefer the following notation, where every *type* $x \in X$ occurring in S is given only once and accompanied by its *repetition number* r_x that captures how often the type occurs in the multiset. The example above is written as: $S = \{2 \cdot \square, 1 \cdot \bigcirc, 3 \cdot \triangle, 1 \cdot \nabla\}$ and we would say the type $\square$ has repetition number 2, the type $\bigcirc$ has repetition number 1 and so on. We write $\square \in S$, $r_{\square} = 2$, $\bigcirc \in S$, $r_{\bigcirc} = 1$ and so on.

As with ordered arrangements, the most important case for unordered arrangements is that all repetition numbers are 1, i.e. $r_x = 1$ for all $x \in S$. Then S is simply a *subset* of X, denoted by $S \subseteq X$.

Definition 1.3. ***k*-Combination:** Let X be a finite set. A *k-combination* of X is an unordered arrangement of k distinct elements from X. We prefer the more standard term *subset* and use "combination" only when we want to emphasize the selection process. The set of all k-subsets of X is denoted by $\binom{X}{k}$ and if $|X| = n$ then we denote

$$\binom{n}{k} := \left|\binom{X}{k}\right|.$$

***k*-Combination of a Multiset:** Let X be a finite set of types and let M be a finite multiset with types in X and repetition numbers $r_1, \ldots, r_{|X|}$. A *k-combination* of M is a multiset with types in X and repetition numbers $s_1, \ldots, s_{|X|}$ such that $s_i \leq r_i$, $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

If for example $M = \{2 \cdot \square, 1 \cdot \bigcirc, 3 \cdot \triangle, 1 \cdot \triangledown\}$, then $T = \{1 \cdot \square, 2 \cdot \triangle\}$ is a 3-combination of the multiset M, but $T' = \{3 \cdot \square, 0 \cdot \triangledown\}$ is not.

***k*-Permutation of a Multiset:** Let M be a finite multiset with set of types X. A *k-permutation* of M is an ordered arrangement of k elements of M where different orderings of elements of the same type are not distinguished. This is a an ordered multiset with types in X and repetition numbers $s_1, \ldots, s_{|X|}$ such that $s_i \leq r_i$, $1 \leq i \leq |X|$, and $\sum_{i=1}^{|X|} s_i = k$.

Note that there might be several elements of the same type compared to a permutation of a set (where each repetition number equals 1). If for example $M = \{2 \cdot \square, 1 \cdot \bigcirc, 3 \cdot \triangle, 1 \cdot \triangledown\}$, then $T = (\square, \triangle, \triangle, \square)$ is a 4-permutation of the multiset M.

Theorem 1.4. *For $0 \leq k \leq n$, we have*

$$P(n,k) = \binom{n}{k} \cdot k! \quad \textit{and therefore} \quad \binom{n}{k} = \frac{n!}{k!(n-k)!}.$$

Proof. To build an element from $P(n,k)$, we first choose k elements from $[n]$, by our definition of $\binom{n}{k}$, there are exactly $\binom{n}{k}$ ways to do so. Then choose an order of the k elements, there are $|P(k,k)| = \frac{k!}{0!} = k!$ ways to do so.

Every element from $P(n,k)$ can be constructed like this in exactly one way so $|P(n,k)| = \binom{n}{k} \cdot k!$ which proves the claim.

Using Theorem 1.2(i) now gives the identity on the right as well. □

The numbers $\binom{n}{k}$ for $0 \leq k \leq n$ are called *binomial coefficients* because these are the coefficients when expanding the n-th power of a sum of two variables:

$$(x+y)^n = \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k}.$$

1.4 Multinomial Coefficients

Definition 1.5. For non-negative integers $k_1, \ldots, k_r$ with $k_1 + \ldots + k_r = n$ the *multinomial coefficients* $\binom{n}{k_1,\ldots,k_r}$ are the coefficients when expanding the n-th power of a sum of r variables. In other words, define them to be the unique numbers satisfying the following identity:

$$(x_1 + \ldots + x_r)^n = \sum_{\substack{k_1,\ldots,k_r \\ k_1+\ldots+k_r=n}} \binom{n}{k_1, \ldots, k_r} x_1^{k_1} x_2^{k_2} \ldots x_r^{k_r}.$$

Example. Verify by multiplying out:

$$\begin{aligned}(x_1 + x_2 + x_3)^4 = \; & 1\cdot(x_1^4 + x_2^4 + x_3^4) \\ & + \; 4\cdot(x_1^3x_2 + x_1^3x_3 + x_1x_2^3 + x_2^3x_3 + x_1x_3^3 + x_2x_3^3) \\ & + \; 6\cdot(x_1^2x_2^2 + x_1^2x_3^2 + x_2^2x_3^2) \\ & + 12\cdot(x_1^2x_2x_3 + x_1x_2^2x_3 + x_1x_2x_3^2).\end{aligned}$$

The coefficient in front of $x_1^2x_2x_3$ is 12, the coefficient in front of $x_1^2x_3^2$ is 6 and the coefficient in front of x_3^4 is 1.

According to the last definition this means:

$$\binom{4}{2,1,1} = 12, \quad \binom{4}{2,0,2} = 6, \quad \binom{4}{0,0,4} = 1.$$

1.4 Multinomial Coefficients

Theorem 1.6 (Multinomial Theorem)**.**
For non-negative integers $k_1, \ldots, k_r$ with $k_1 + \ldots + k_r = n$ we have:

$$\binom{n}{k_1, \ldots, k_r} = \binom{n}{k_1} \cdot \binom{n-k_1}{k_2} \cdot \ldots \cdot \binom{n-k_1-\ldots-k_{r-1}}{k_r} = \frac{n!}{k_1! \cdot k_2! \cdot \ldots \cdot k_r!}.$$

Proof. The naive way to multiply out $(x_1 + \ldots + x_r)^n$ would be:

$$(x_1 + \ldots + x_r)^n = \underbrace{\sum_{i_1=1}^{r} \sum_{i_2=1}^{r} \ldots \sum_{i_n=1}^{r}}_{n \text{ summation signs}} x_{i_1} x_{i_2} \ldots x_{i_n}.$$

The monomial $x_{i_1} x_{i_2} \ldots x_{i_n}$ is equal to $x_1^{k_1} \ldots x_r^{k_r}$ if from the indices $\{i_1, \ldots, i_n\}$ exactly k_1 are equal to 1, k_2 equal to 2 and so on.

We count the number of assignments of values to the indices $\{i_1, \ldots, i_n\}$ satisfying this.

Choose k_1 indices to be equal to 1: There are $\binom{n}{k_1}$ ways to do so.

Choose k_2 indices to be equal to 2: There are $n - k_1$ indices left to choose from, so there are $\binom{n-k_1}{k_2}$ ways to choose k_2 of them.

Choose k_j indices to be equal to j ($j \in [r]$): There $n - k_1 - \ldots - k_{j-1}$ indices left to choose from, so there are $\binom{n-k_1-\ldots-k_{j-1}}{k_j}$ ways to choose k_j of them.

Hence the multinomial coefficient (i.e. the coefficient of $x_1^{k_1} x_2^{k_2} \ldots x_r^{k_r}$) is

$$\binom{n}{k_1, \ldots, k_r} = \binom{n}{k_1} \cdot \binom{n-k_1}{k_2} \cdot \ldots \cdot \binom{n-k_1-k_2-\ldots-k_{r-1}}{k_r}$$

and the first identity is proved. Now use Theorem 1.4 to rewrite the binomial coefficients and obtain

$$\frac{n!}{k_1!(n-k_1)!} \cdot \frac{(n-k_1)!}{k_2!(n-k_1-k_2)!} \cdot \ldots \cdot \frac{(n-k_1-\ldots-k_{r-1})!}{k_r!(n-k_1-\ldots-k_r)!}$$

Conveniently, many of the factorial terms cancel out, like this:

$$\frac{n!}{k_1!\cancel{(n-k_1)!}} \cdot \frac{\cancel{(n-k_1)!}}{k_2!\cancel{(n-k_1-k_2)!}} \cdot \ldots \cdot \frac{\cancel{(n-k_1-\ldots-k_{r-1})!}}{k_r!(n-k_1-\ldots-k_r)!}.$$

Also, $(n - k_1 - \ldots - k_r)! = (n-n)! = 0! = 1$ so we end up with:

$$\frac{n!}{k_1! \cdot k_2! \cdot \ldots \cdot k_r!}.$$

□

Example. $\binom{5}{3,0,2} = \frac{5!}{3!0!2!} = \frac{5\cdot 4\cdot \cancel{3}\cdot \cancel{2}\cdot \cancel{1}}{\cancel{3}\cdot \cancel{2}\cdot \cancel{1}\cdot 2\cdot 1} = 10.$

Note that the last Theorem establishes that binomial coefficients are special cases of multinomial coefficients. We have for $0 \leq k \leq n$:

$$\binom{n}{k} = \frac{n!}{k!(n-k)!} = \binom{n}{k, n-k}.$$

We extend the definition of binomial and multinomial coefficients by setting $\binom{n}{k_1,\ldots,k_r} = 0$ if $k_i = -1$ for some i, and $\binom{n}{-1} = \binom{n}{n+1} = 0$. This makes stating the following lemma more convenient.

Lemma 1.7 (Pascal's Formula)**.**
If $n \geq 1$ and $0 \leq k \leq n$, we have

$$\binom{n}{k} = \binom{n-1}{k} + \binom{n-1}{k-1}.$$

More generally, for $n \geq 1$ and $k_1, \ldots, k_r \geq 0$ with $k_1 + \ldots + k_r = n$, we have

$$\binom{n}{k_1, \ldots, k_r} = \sum_{i=1}^{r} \binom{n-1}{k_1, \ldots, k_i - 1, \ldots, k_r}.$$

Proof. Note first, that in the case of binomial coefficients, the claim can be rewritten as:

$$\binom{n}{k, n-k} = \binom{n-1}{k, n-k-1} + \binom{n-1}{k-1, n-k}$$

so it really is a special case of the second claim, which we prove now. By definition of multinomial coefficients we have the identity:

$$(x_1 + \ldots + x_r)^n = \sum_{\substack{k_1, \ldots, k_r \geq 0 \\ k_1 + \ldots + k_r = n}} \binom{n}{k_1, \ldots, k_r} x_1^{k_1} \ldots x_r^{k_r}.$$

Exploring a different way to expand $(x_1 + \ldots + x_r)^n$, we obtain:

$$\begin{aligned}
(x_1 + \ldots + x_r)^n &= (x_1 + \ldots + x_r) \cdot (x_1 + \ldots + x_r)^{n-1} \\
&= (x_1 + \ldots + x_r) \cdot \sum_{\substack{k_1', \ldots, k_r' \geq 0 \\ k_1' + \ldots + k_r' = n-1}} \binom{n-1}{k_1', \ldots, k_r'} x_1^{k_1'} \ldots x_r^{k_r'} \\
&= \sum_{i=1}^{r} \sum_{\substack{k_1', \ldots, k_r' \geq 0 \\ k_1' + \ldots + k_r' = n-1}} \binom{n-1}{k_1', \ldots, k_r'} x_1^{k_1'} \ldots x_i^{k_i'+1} \ldots x_r^{k_r'}.
\end{aligned}$$

By substituting indices as $k_i := k_i' + 1$ and $k_j := k_j'$ (for $j \neq i$) this is equal to:

$$(x_1 + \ldots + x_r)^n = \sum_{i=1}^{r} \sum_{\substack{k_1,\ldots,k_r \geq 0 \\ k_i - 1 \geq 0 \\ k_1 + \ldots + k_r = n}} \binom{n-1}{k_1, \ldots, k_i - 1 \ldots, k_r} x_1^{k_1} \ldots x_r^{k_r}.$$

Note that the condition $k_i - 1 \geq 0$ is not needed, because for the summands with $k_i - 1 = -1$ we defined the coefficient under the sum to be 0. We remove it and swap the summation signs, ending up with:

$$(x_1 + \ldots + x_r)^n = \sum_{\substack{k_1,\ldots,k_r \geq 0 \\ k_1 + \ldots + k_r = n}} \sum_{i=1}^{r} \binom{n-1}{k_1, \ldots, k_i - 1 \ldots, k_r} x_1^{k_1} \ldots x_r^{k_r}$$

Now we have two ways to write the coefficients of $(x_1 + \ldots + x_r)^n$ and therefore the identity:

$$\binom{n}{k_1, \ldots, k_r} = \sum_{i=1}^{r} \binom{n-1}{k_1, \ldots, k_i - 1 \ldots, k_r}$$

as claimed. □

1.4 Multinomial Coefficients

You may already know Pascal's Triangle. It is a way to arrange binomial coefficients in the plane as shown in Figure 9. It is possible to do something similar in the case of multinomial coefficients, however, when drawing all coefficients of the form $\binom{n}{k_1 \ldots, k_r}$, the drawing will be r-dimensional. For $r = 3$, a "Pascal Pyramid" is given in Figure 10.

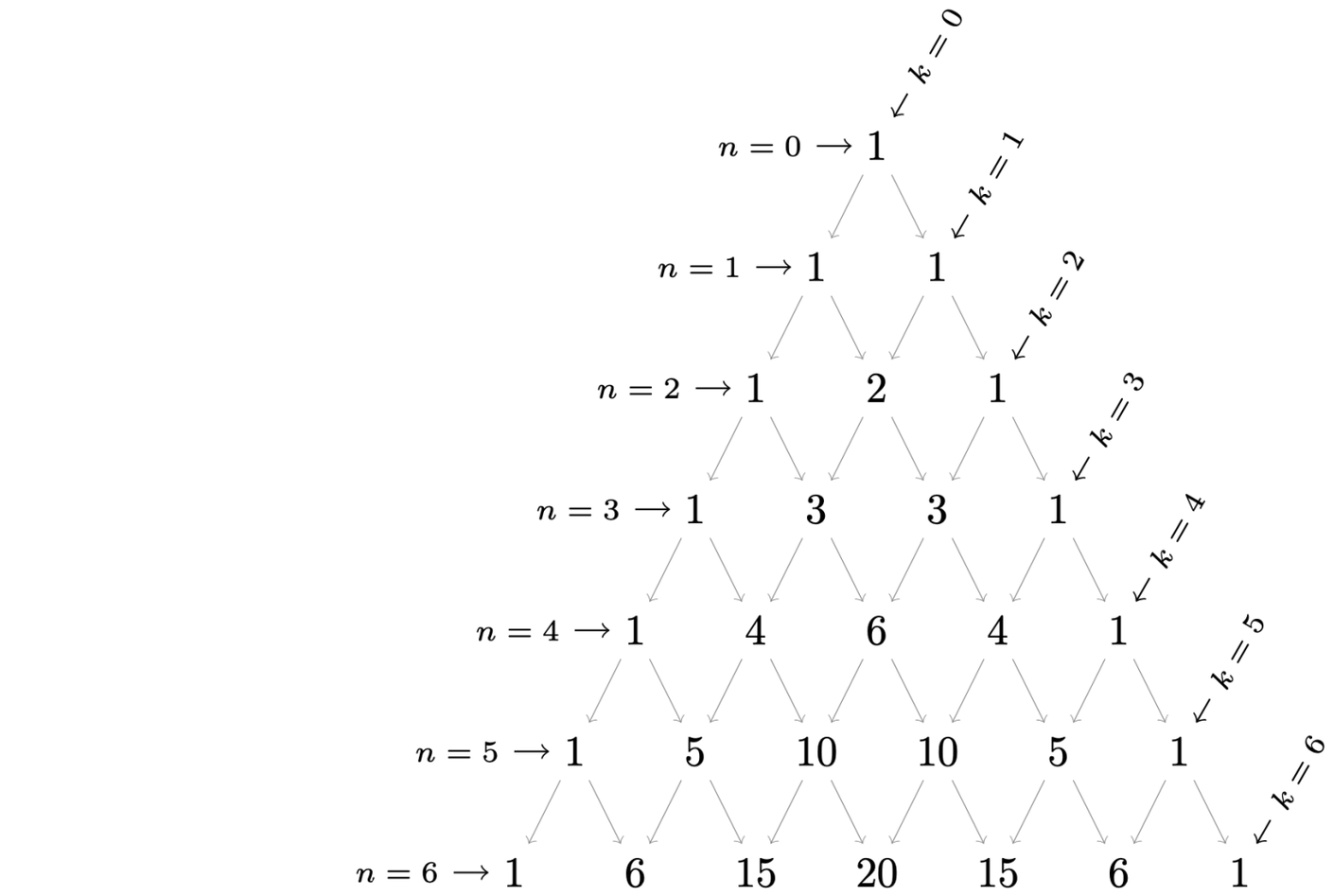
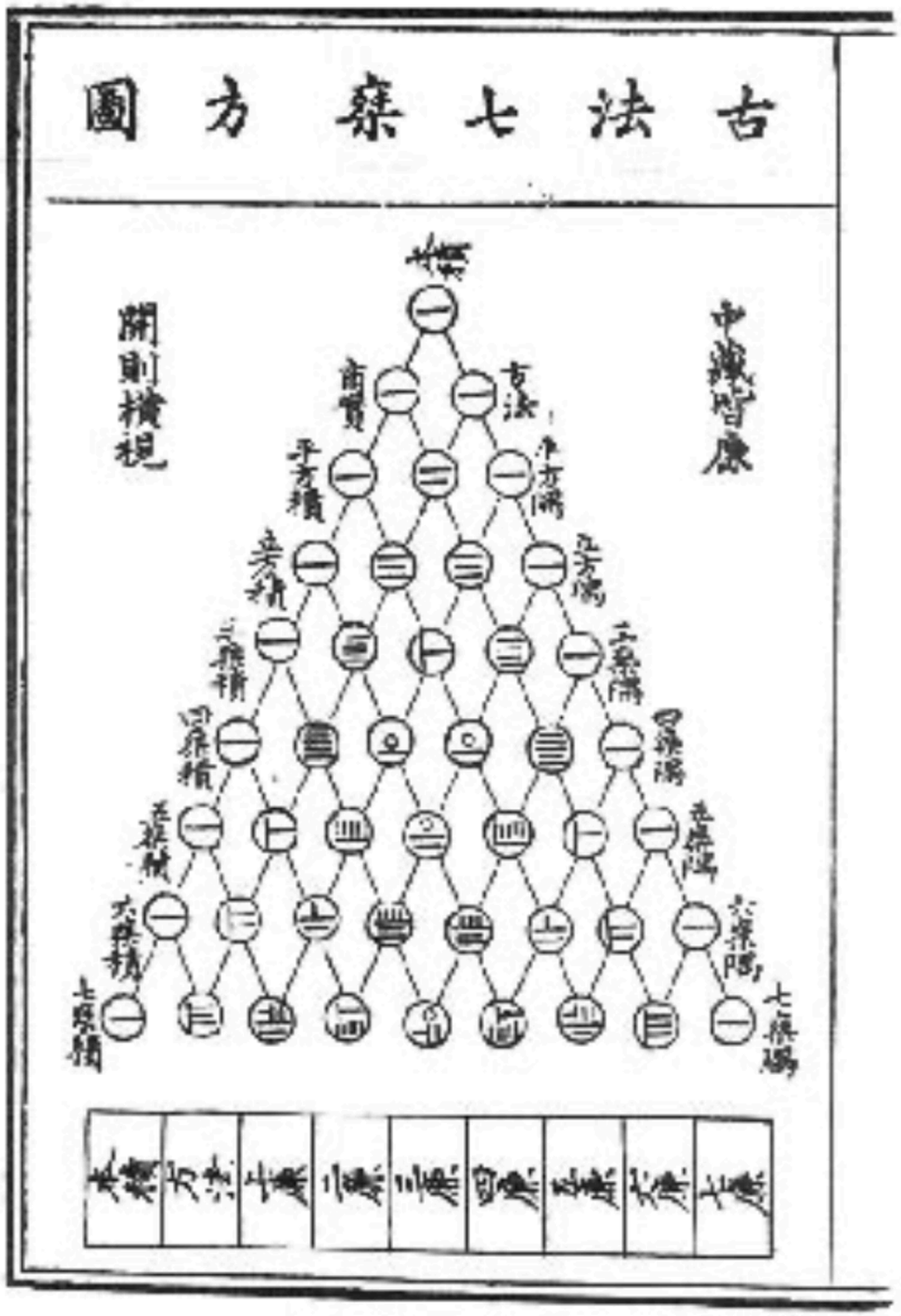


Figure 9: Arrangement of the binomial coefficients $\binom{n}{k}$. Lemma 1.7 shows that the number $\binom{n}{k}$ is obtained as the sum of the two numbers $\binom{n-1}{k-1}$ and $\binom{n-1}{k}$ directly above it.

1.4 Multinomial Coefficients

Although we call this Pascal's Triangle, Pascal was not the first person to write it down. Below is a version due to Chu-Shi-Chieh (Zhu Shijie), taken from work of Yang Hui, in his book *Ssu Yuan Yu¨ Chien*, dated 1303. Jia Xian knew it about 250 years earlier. Other people who knew about it at roughly the same time include Halayudha in India, and Al-Karaji and Omar Khayyam in Iran.

We don't know who invented it!



1.4 Multinomial Coefficients

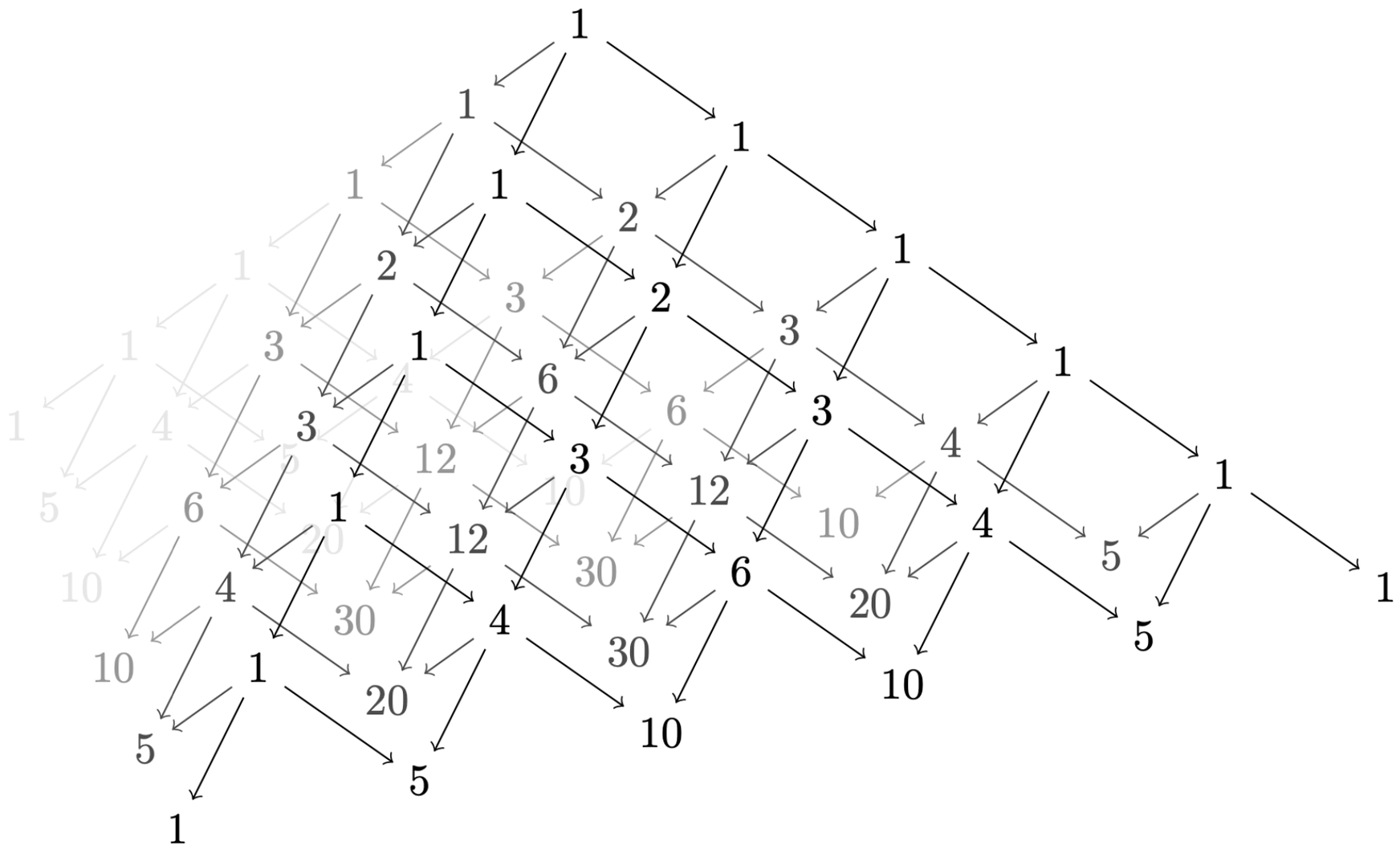


Figure 10: Arrangement of the numbers $\binom{n}{k_1,k_2,k_3}$ with $k_1 + k_2 + k_3 = n$ in a pyramid. To make the picture less messy, the numbers in the back (with large k_3) are faded out. The three flanks of the pyramid are pascal triangles, one of which is the black triangle in the front with numbers $\binom{n}{k_1,k_2,0}$. The first number not on one of the flanks is $\binom{3}{1,1,1} = 6$.

1.4 Multinomial Coefficients

Example. How many 6-permutations are there of the multiset $\{\bigcirc, \triangle, \triangle, \square, \square, \square\}$? Trying to list them all $((\bigcirc, \triangle, \triangle, \square, \square, \square), (\triangle, \bigcirc, \triangle, \square, \square, \square),), (\triangle, \triangle, \bigcirc, \square, \square, \square), \ldots)$ would be tedious. Well, there are 6 possible positions for $\bigcirc$. Then there are 5 positions left, 2 of which need to contain $\triangle$. The three $\square$ need to go to the three remaining positions, so there is just one choice left for them. This means there are $6 \cdot \binom{5}{2} \cdot 1 = 60$ arrangements in total. This is equal to $\binom{6}{1}\binom{5}{2}\binom{3}{3} = \binom{6}{1,2,3}$, which is no coincidence:

Theorem 1.8. *Let S be a finite multiset with k different types and repetition numbers $r_1, r_2, \ldots r_k$. Let the* size *of S be $n = r_1 + r_2 + \cdots + r_k$. Then the number of n-permutations of S equals*

$$\binom{n}{r_1, \ldots, r_k}.$$

Proof. Label the k types as $a_1, \ldots, a_k$. In an n-permutation there are n positions that need to be assigned a type. First choose the r_1 positions for the first type, there are $\binom{n}{r_1}$ ways to do so. Then assign r_2 positions for the second type, out of the $n - r_1$ positions that are still left to choose. That amounts to $\binom{n-r_1}{r_2}$ choices. Continuing like this, the total number of choices will be:

$$\binom{n}{r_1} \cdot \binom{n-r_1}{r_2} \cdot \ldots \cdot \binom{n-r_1-r_2-\ldots-r_{k-1}}{r_k} \overset{\text{Thm 1.6}}{=} \binom{n}{r_1, r_2, \ldots, r_k}.$$

□

1.4 Multinomial Coefficients

We have seen before how the coefficient $\binom{n}{k} = \binom{n}{k,n-k}$ counts the number of k-combinations of $[n]$ (i.e. number of subsets of $[n]$). Now we learned, that it also counts the number of n-permutations of a multiset with two types and repetition number k and $n-k$. How come ordered and unordered things are counted by the same number? We demystify this by finding a natural bijection:

Consider the multiset $M := \{k \cdot ✓, (n-k) \cdot ✗\}$ with types ✓ and ✗ (the "chosen" type and the "unchosen" type). Now associate with an n-permutation of M the set of positions that contain ✓, for instance with $n = 5$, $k = 2$ and the permutation $(✓, ✗, ✗, ✓, ✗)$, the corresponding set would be $\{1, 4\} \subset [5]$ (since the first and fourth position received ✓). It is easy to see that every n-permutation of M corresponds to a unique k element subset of $[n]$ and vice versa.

Note that so far we only considered n-permutations of multisets of size n. What about general r-permutations of multisets of size n? For example, the number of 2 permutations of the multiset $\{□, △, △, △, ○\}$ is 7, since there is:

$$□△, \quad □○, \quad △□, \quad △△, \quad △○, \quad ○□, \quad ○□.$$

Note that ○○ and □□ is not possible, since we have only of one copy of ○ and □ at our disposal. The weird number 7 already suggests that general r-permutations of n element multisets may not be as easy to count. Indeed, there is no simple formula as in Theorem 1.8 but we will see an answer using *the principle of inclusion and exclusion* later.

1.4 Multinomial Coefficients

There is a special case other than $r = n$ that we can handle, though: If all repetition numbers r_i of a multiset with k types are bigger or equal than r, for instance when considering 2-permutations of $M := \{\square, \square, \triangle, \triangle, \triangle, \bigcirc, \bigcirc\}$, then those repetition numbers do not actually impose a restriction, since we will never run out of copies of any type. We would sometimes sloppily write $M = \{\infty \cdot \square, \infty \cdot \triangle, \infty \cdot \bigcirc\}$ where the infinity sign indicates that there are "many" copies of the corresponding elements. The number of r-permutations of M is then equal to k^r, just choose one type of the k types for each of the r positions.

After permutations of multisets, we now consider combinations.

Example. Say you are told to bring two pieces of fruit from the supermarket and they got 🍌,🍎 and 🍐 (large quantities of each). How many choices do you have? Well, there is: {🍌, 🍌}, {🍌, 🍎}, {🍌, 🍐}, {🍎, 🍎}, {🍎, 🍐}, {🍐, 🍐}, so six combinations. Note that bringing a 🍐 and an 🍎 is the same as bringing an 🍎 and a 🍐 (your selection is not ordered), so this option is counted only once.

We now determine the number of combinations for arbitrary number of types and number of elements to choose.

1.4 Multinomial Coefficients

Theorem 1.9. *Let $r, k \in \mathbb{N}$ and let S be a multiset with k types and large repetition numbers (each $r_1, \ldots, r_k$ is at least r), then the number of r-combinations of S equals*

$$\binom{k+r-1}{r}.$$

Proof. For clarity, we do the proof alongside an example. Let the types be $a_1, a_2, \ldots, a_k$, for instance $k = 4$ and $a_1 =$ 🍌, $a_2 =$ 🍎, $a_3 =$ 🍐, $a_4 =$ 🍊. Then imagine the r-combinations laid-out linearly, first all elements of type a_1 then all of type a_2 and so on. In our example this could be

🍌🍌🍐🍐🍊 for the combination $\{2 \cdot$ 🍌$, 2 \cdot$ 🍐$, 1 \cdot$ 🍊$\}$.

Now for each $i \in [k-1]$, draw a delimiter between types a_i and a_{i+1}, in our example:

🍌🍌 | | 🍐🍐 | 🍊.

Note that, since we have no elements of type $a_2 =$ 🍎, there are two delimiters directly after one another. Given these delimiters, drawing the elements is actually redundant, just replacing every element with "•" yields:

• • | | • • | •.

The original k-combination can be reconstructed from this sequence of | and •: Just replace every • with a_1 until the first occurrence of |, then use a_2 until the next |, then a_3 and so on. So every $(r+k-1)$-permutation of $T := \{(k-1)\cdot|, r\cdot\bullet\}$ corresponds to an r-combination of S and vice versa.

We know how to count the former, by Theorem 1.8 the number of $(r+k-1)$-permutations of T is $\binom{r+k-1}{r,k-1} = \binom{r+k-1}{r}$. □

Counting r-combinations of multisets where repetition numbers $r_1, \ldots, r_k$ may be smaller than r is more difficult. We will see later how the *inclusion-exclusion principle* provides an answer in this case.

1.4 Multinomial Coefficients

Theorem

For $n \geq 0$,

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \ldots + \binom{n}{n} = 2^n.$$

Proof. (I) mathematical reasoning; (II) combinatorics

1.4 Multinomial Coefficients

Q1. Calculate the values of $\binom{-1}{m}$ and $\binom{-2}{m}$

$$\binom{-1}{m} = \frac{(-1)^{\underline{m}}}{m!} = \frac{(-1)(-2)\dots(-m)}{m!} = (-1)^m,$$

$$\binom{-2}{m} = \frac{(-2)^{\underline{m}}}{m!} = \frac{(-2)(-3)\dots(-m-1)}{m!} = (-1)^m(m+1),$$

Q2. Calculate the values of $\binom{1/2}{m}$

$$\begin{aligned}
\binom{1/2}{m} &= \frac{(\frac{1}{2})(\frac{-1}{2})(\frac{-3}{2})\dots(\frac{1}{2}-m+1)}{m!} \\
&= \frac{(-1)(-3)\dots(3-2m)}{2^m m!} \\
&= \frac{(-1)^{m-1}(1)(3)\dots(2m-3)}{2^m m!} \\
&= \frac{(-1)^{m-1}(2)(3)\dots(2m-3)(2m-2)}{2^m m!(2)(4)\dots(2m-2)} \\
&= \frac{(-1)^{m-1}}{2^m 2^{m-1}} \times \frac{(2m-2)!}{m!(m-1)!} \\
&= \frac{(-1)^{m-1}}{2^{2m-1}} \times \frac{\binom{2m-2}{m-1}}{m}
\end{aligned}$$

1.4 Multinomial Coefficients

Q2. Calculate the values of $\binom{1/2}{m}$

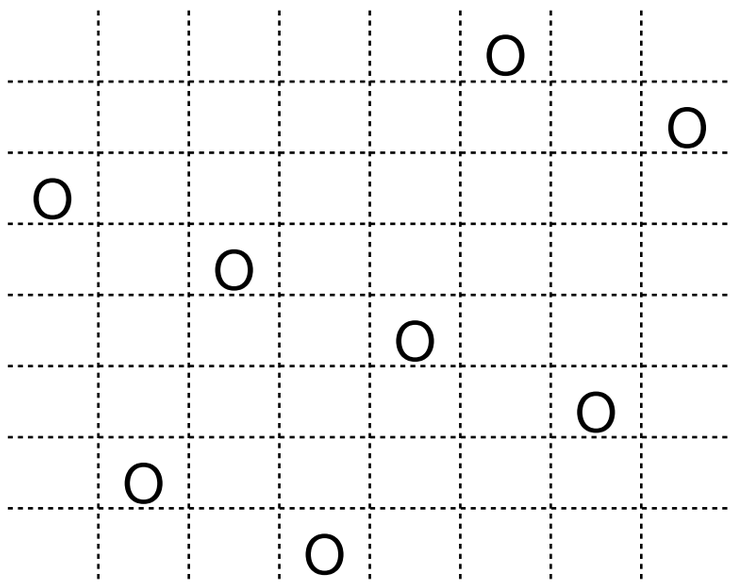
$$\begin{aligned}
\binom{1/2}{m} &= \frac{(\frac{1}{2})(\frac{-1}{2})(\frac{-3}{2})\dots(\frac{1}{2}-m+1)}{m!} \\
&= \frac{(-1)(-3)\dots(3-2m)}{2^m m!} \\
&= \frac{(-1)^{m-1}(1)(3)\dots(2m-3)}{2^m m!} \\
&= \frac{(-1)^{m-1}(2)(3)\dots(2m-3)(2m-2)}{2^m m!(2)(4)\dots(2m-2)} \\
&= \frac{(-1)^{m-1}}{2^m 2^{m-1}} \times \frac{(2m-2)!}{m!(m-1)!} \\
&= \frac{(-1)^{m-1}}{2^{2m-1}} \times \boxed{\frac{\binom{2m-2}{m-1}}{m}}
\end{aligned}$$

Catalan number

1.4 Multinomial Coefficients

Game of Chess: Two rooks can attack one another if and only if they lie in the same row or the same column of the chessboard.

An example of 8 non attacking rooks on a 8-by-8 board is



The rooks must occupy eight squares with coordinates

$$(1,j_1), (2,j_2), \ldots, (8,j_8)$$

$j_1, j_2, \ldots, j_8$ is a permutation of 1,2,…,8. Therefore, there are 8! approaches.

1.4 Multinomial Coefficients

Game of Chess: Two rooks can attack one another if and only if they lie in the same row or the same column of the chessboard.

The previous slide consider the case that all rooks are the same.

What is the case that all rooks are different? They are of different colors.

$$8!8!$$

Now we consider the case that $\{1 \cdot R,\ 3 \cdot B,\ 4 \cdot Y\}$

By Theorem 2.4.2, the number of permutations of this multiset equals $\dfrac{8!}{1!3!4!}$

The number of ways to place these rooks on an 8-by-8 board so that no rook can attack another equals

$$\frac{8!}{1!3!4!}8!.$$

1.4 Multinomial Coefficients

Game of Chess: Two rooks can attack one another if and only if they lie in the same row or the same column of the chessboard.

Theorem

There are n rooks of k colors with n_1 rooks of the first color, n_2 rooks of the second color, ..., and n_k rooks of the k-th color. The number of ways to arrange these rooks on an n-by-n board so that no rook can attack another equals

$$n!\frac{n!}{n_1!n_2!\ldots n_k!} = \frac{(n!)^2}{n_1!n_2!\ldots n_k!}.$$

Thanks for your attention!

Any question?